

Your Paper Title Here

Your Name(s)

Date

Abstract

Write your abstract here.

Keywords

Keyword 1, Keyword 2, Keyword 3, Keyword 4

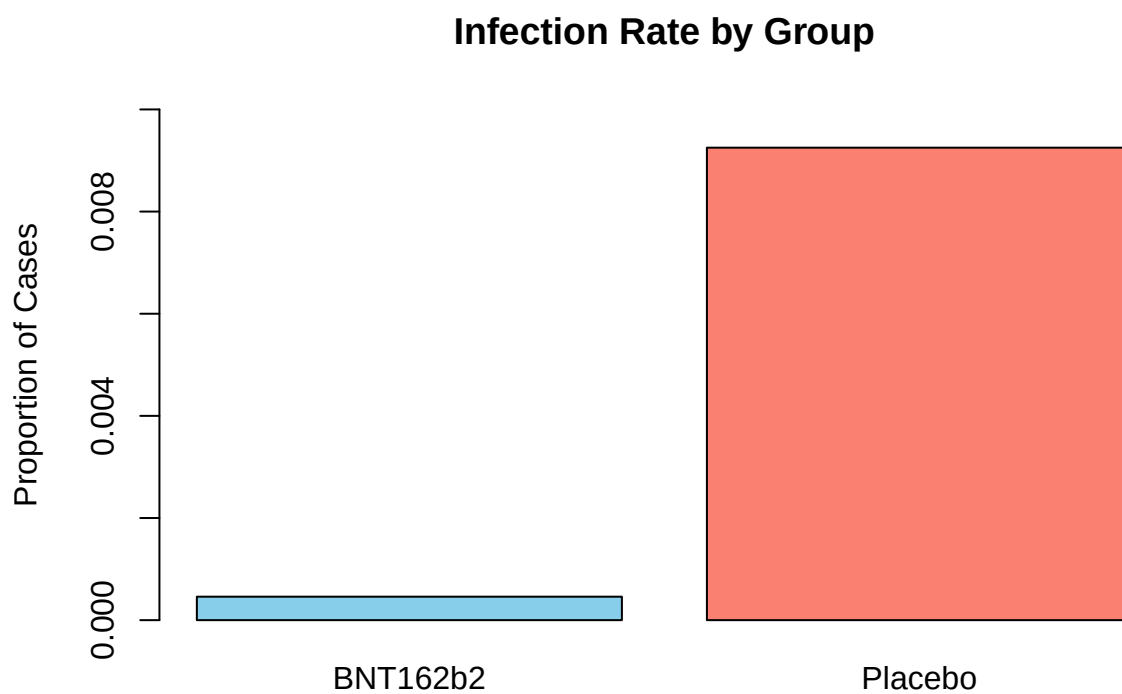
Introduction / Background

The December 2020 Emergency Use Authorization (EUA) of the BNT162b2 vaccine marked a pivotal shift in the global effort to mitigate the SARS-CoV-2 pandemic. Developed through a collaboration between Pfizer and BioNTech, this two-dose regimen represented a departure from traditional vaccine technologies. As noted in contemporary scientific literature, while viral vector approaches were utilized by manufacturers like Janssen and AstraZeneca, BNT162b2 utilized lipid nanoparticle-formulated nucleoside-modified RNA (mRNA) to induce an immune response against the viral spike protein. To evaluate the safety and clinical impact of this technology, a rigorous, multinational, placebo-controlled efficacy study was conducted and published by Polack et al. (2020). This trial followed a 1:1 randomization ratio, assigning over 34,000 participants to either the vaccine or placebo groups. The study was designed to reach statistical maturity upon the occurrence of 170 laboratory-confirmed COVID-19 cases in participants without prior evidence of infection. The resulting data demonstrated a significant disparity in outcomes. Of the 170 confirmed cases, 8 occurred in the BNT162b2 cohort ($n = 17,411$), while 162 occurred in the placebo cohort ($n = 17,511$). This results in an observed infection rate of approximately 0.046% for the vaccine group compared to 0.925% for the placebo group.

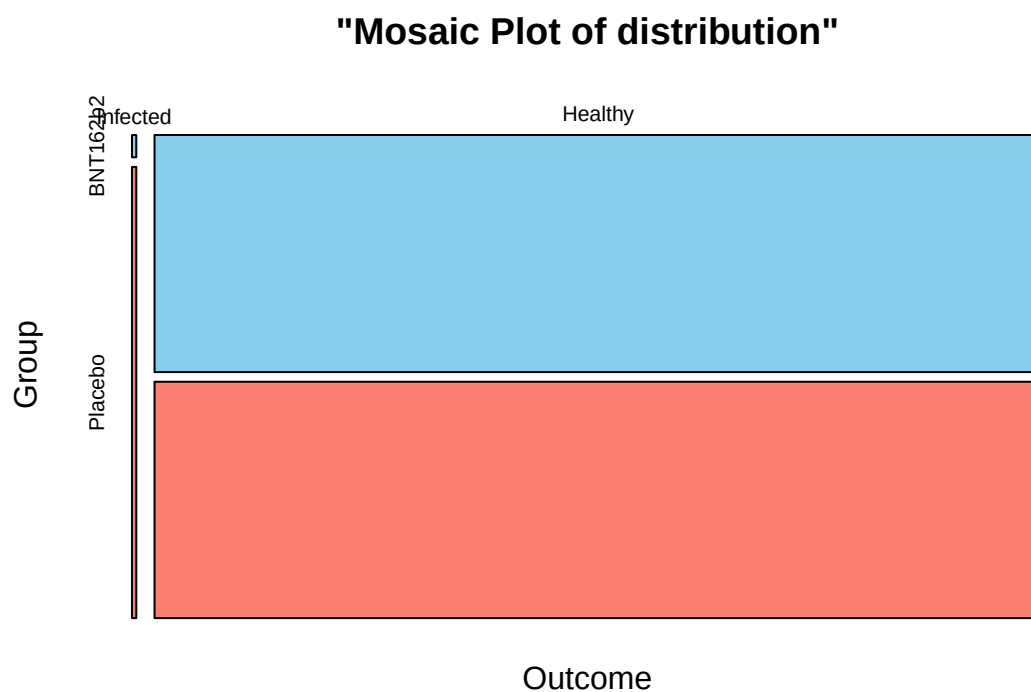
```
trial_matrix <- matrix(c(8, 17411-8, 162, 17511-162),
                      nrow = 2,
                      dimnames = list(Outcome = c("Infected", "Healthy"),
                                       Group = c("BNT162b2", "Placebo")))

rates <- trial_matrix["Infected", ] / colSums(trial_matrix)

barplot(rates,
        main="Infection Rate by Group",
        ylab="Proportion of Cases",
        col=c("skyblue", "salmon"),
        ylim=c(0, 0.01))
```



```
array <- as.array(trial_matrix)
mosaicplot(array, main=deparse1(substitute("Mosaic Plot of distribution")), sub = NULL, xlab =
  sort = NULL, off = NULL, dir = NULL,
  color = c("skyblue","salmon"), shade = FALSE, margin = NULL,
  cex.axis = 0.66, las = par("las"), border = NULL)
```



This analysis seeks to evaluate the efficacy of the vaccine based on these findings. Specifically, we test whether the vaccine efficacy (ψ) is significantly greater than the 30% threshold required for regulatory approval. We formulate our research goal through the hypotheses $H_0 : \psi = 0.3$ versus $H_1 : \psi > 0.3$.

Statistical Methods

Model

Describe the statistical model used.

Likelihood Inference

Detail the likelihood approach.

Bayesian Inference

The Bayesian approach requires the specification of a prior distribution for π that represents our state of knowledge before observing the trial data. We adopt a conjugate Beta prior, $\pi \sim \text{Beta}(a, b)$, which is mathematically advantageous as it ensures the posterior distribution will also belong to the Beta family. To determine the values of the hyperparameters a and b , we perform a prior elicitation based on a “pessimistic” perspective of vaccine performance. We assume a prior median for π of 0.5, reflecting an initial belief that the vaccine and placebo groups are equally likely to result in an infection. Furthermore, we reflect regulatory skepticism by setting only a 5% prior probability that the vaccine efficacy meets the 30% threshold ($P(\psi \geq 0.30) = 0.05$). Converting this efficacy threshold to the π scale using the transformation $\pi = \frac{1-\psi}{2-\psi}$, we target a distribution where the 5th percentile is approximately 0.4118. Solving these constraints yields the parameters:

$$a = 43.04, \quad b = 43.04$$

The likelihood of observing t infections in the vaccine group out of n total cases is given by the binomial probability mass function:

$$f(t|\pi) = \binom{n}{t} \pi^t (1 - \pi)^{n-t}$$

Applying Bayes’ Theorem, the posterior distribution $f(\pi|t)$ is proportional to the product of the likelihood and the prior. Due to the conjugacy of the Beta-binomial model, the posterior distribution for π is updated by adding the observed successes and failures to the prior parameters:

$$\pi|t \sim \text{Beta}(a + t, b + n - t)$$

With our observed data of $t = 8$ and $n = 170$, the posterior becomes $\pi|t \sim \text{Beta}(51.04, 205.04)$. Following the calculation of this distribution, we transform the results back to the ψ scale. This allows for the determination of the posterior median for vaccine efficacy and a 95% Highest Posterior Density Interval (HPDI) to quantify uncertainty. Finally, the posterior probability $P(\psi > 0.30|t)$ is calculated to evaluate the strength of evidence in favor of the research hypothesis.

```

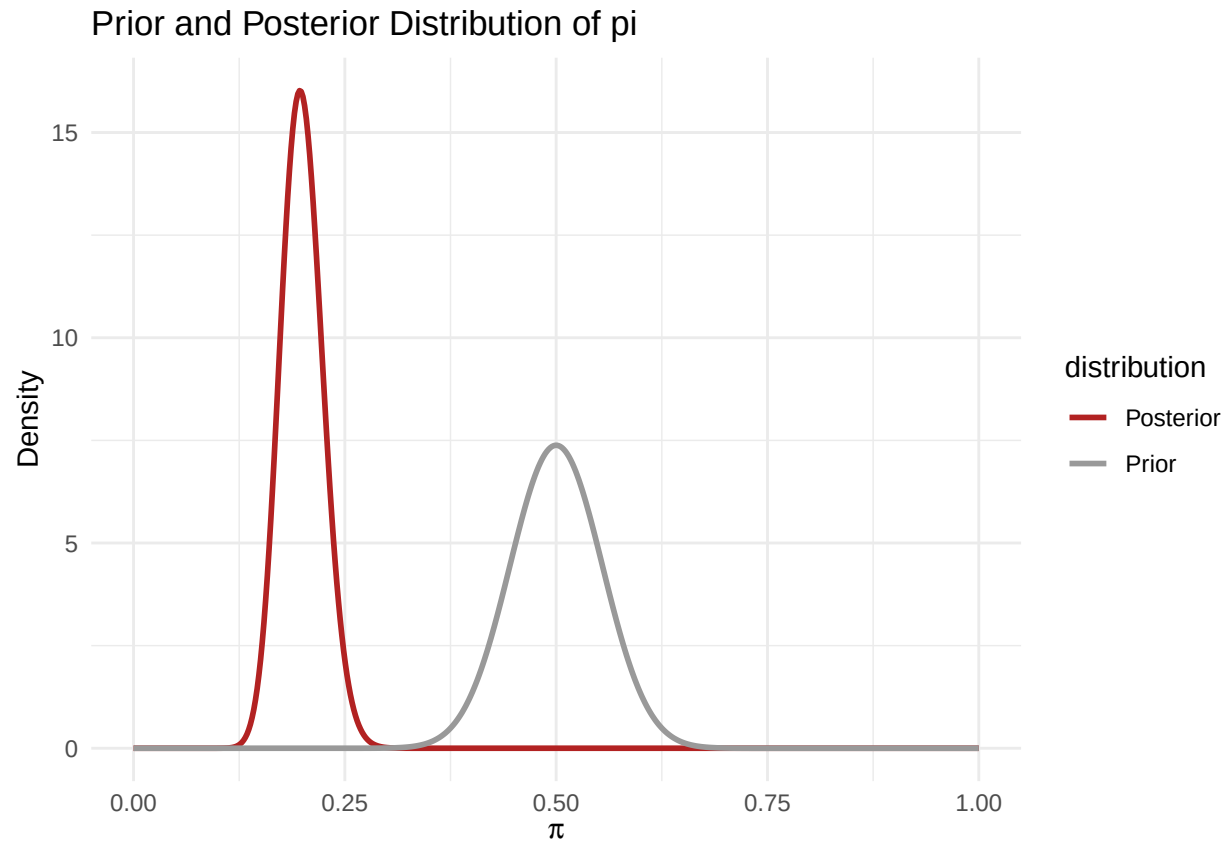
library(ggplot2)

# Define parameters
a_prior <- 43.04
b_prior <- 43.04
t_obs <- 8
n_total <- 170
a_post <- a_prior + t_obs
b_post <- b_prior + (n_total - t_obs)

# Generate data for plotting
x <- seq(0, 1, length.out = 500)
df <- data.frame(
  pi = rep(x, 2),
  pdf = c(dbeta(x, a_prior, b_prior), dbeta(x, a_post, b_post)),
  distribution = rep(c("Prior", "Posterior"), each = 500)
)

# Plot
ggplot(df, aes(x = pi, y = pdf, color = distribution)) +
  geom_line(linewidth = 1) +
  labs(title = "Prior and Posterior Distribution of pi",
       x = expression(pi),
       y = "Density") +
  theme_minimal() +
  scale_color_manual(values = c("Prior" = "gray60", "Posterior" = "firebrick"))

```



Results

Present your findings.

Discussion / Conclusion

Discuss / conclude here.

Bibliography

Brown, B. (2024). *Lecture Title*. Lecture slides, Course Name, University Name.

Doe, J. (2020). Title of the Paper. *Journal Name*, 12(3), 45-67.

Last, F., & Last, F. (2025). *Book Title*. Publisher.

Smith, A., & Johnson, C. (2023). *Title of the Online Article*. Retrieved from <https://www.example>.

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Appendix

Code

```
# You can reference your code in the appendix (sample here).
```

Proofs

If applicable, include detailed mathematical derivations or additional theoretical explanations.